Long-term mortality in patients with myocardial injury after non-cardiac surgery

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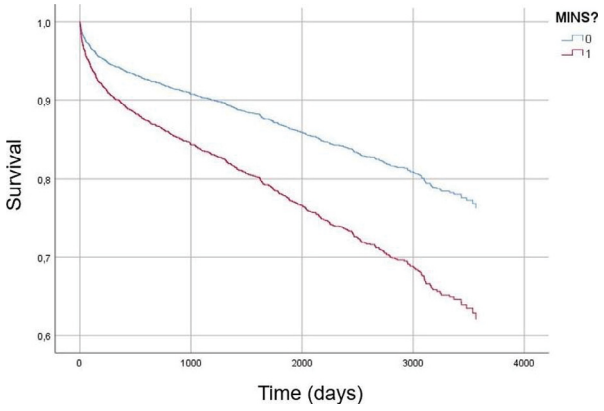
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Background: Myocardial injury that occurs after non-cardiac surgery (MINS) increases mortality in 30 days. However, few studies have assessed the impact of MINS occurrence on long-term mortality.

Methods: Retrospective study including all non-cardiac surgery patients admitted to an postoperative care unit who stayed at least one night in this unit, and had at least one measurement of high-sensitive cardiac troponin. Patients undergoing coronary angiography and patients with an underlying disease with a poor prognosis were excluded. The study period was from January/2011 to December/2019, ensuring a minimum follow-up of 2 years. Clinical characteristics and the occurrence of MINS were assessed between deaths and survivors using the chi-square test and Student's t-test. Variables with $p<0.01$ in the univariate model were included in the Cox regression model to identify predictor variables.

Results: 2532 patients were included, with a mean follow-up of 8.62±2.72 years, mean age = 63.23±16.5 years and 53.6% women. The prevalence of MINS was 12.5% (317 patients). There were 620 deaths (24.5%). In the univariate analysis, the following variables were associated with death: MINS (OR 2.65; $p<0.001$); previous heart failure (OR 3.87; $p<0.001$); chronic renal failure (OR 5.0; $p<0.001$), arterial hypertension (OR 1.83; $p<0.001$), previous myocardial infarction (OR 2.57; $p<0.001$), atrial fibrillation (OR 2.59; $p<0.001$), dementia (OR 8.15; $p<0.001$) and age (60±16 x 73±13, $p<0.001$). These variables were included in Cox's regression model and the predictive variables of all-cause death were: MINS (HR 1.76; CI95% 1.44–2.15), previous heart failure (HR 1.85; CI95% 1.18–2.90), previous myocardial infarction (HR 1.46; CI95% 1.17–1.83); dementia (HR 2.85; CI95% 2.18–3.75) and age (HR 1.05; CI95% 1.04–1.06).

Conclusion: In patients undergoing non-cardiac surgery, the increase in high-sensitive troponin increased the risk of death from all causes in the long term.



Survival curve

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